MODEL COMPARISON SUMMARY

OVERVIEW:

In this project, two machine learning models were developed and evaluated for customer churn prediction:

- Logistic Regression (Baseline Model)
- XGBoost (Advanced Model)

Both models were trained on the same preprocessed dataset, with appropriate techniques applied to handle class imbalance. The models were evaluated using multiple performance metrics, including Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

HANDLING CLASS IMBALANCE:

To address the imbalanced nature of the dataset (~80% non-churn, ~20% churn), the following industry-oriented techniques were used:

- Logistic Regression → class_weight = 'balanced'
- XGBoost → scale_pos_weight

These approaches ensure that the minority class (churn) is given higher importance during training without modifying the original dataset.

MODEL PERFORMANCE:

Logistic Regression (Baseline)

- Accuracy: 0.7420
- F1 Score: 0.5000
- Recall: 0.6339
- Precision: 0.4128
- ROC-AUC: 0.7791

OBSERVATION:

Logistic Regression provides moderate performance but struggles to capture complex patterns in the data, resulting in lower recall and F1-score.

Logistic Regression – After Threshold Tuning

- Accuracy: 0.3040
- F1 Score: 0.3650
- Recall: 0.9828
- Precision: 0.2241
- ROC-AUC: 0.7791

After tuning:

- Recall became extremely high (**98%**)
- Precision dropped significantly (**22%**)
- Accuracy dropped heavily

OBSERVATION:

Logistic Regression tuning leads to **over-prediction of churn**, making the model impractical for real-world usage.So tuning for baseline model is not possible

XGBoost (Before Threshold Tuning)

- Accuracy: 0.8000
- F1 Score: 0.6000
- Recall: 0.7371
- ROC-AUC: 0.8655

OBSERVATION:

XGBoost significantly improves performance compared to Logistic Regression, especially in recall and ROC-AUC, indicating better class separation.

XGBoost (After Threshold Tuning)

- Optimal Threshold: 0.42
- Accuracy: 0.7620
- F1 Score: 0.5795
- Recall: 0.8059
- Precision: 0.4524
- ROC-AUC: 0.8655

OBSERVATION:

Threshold tuning improves recall further, allowing the model to detect more churn customers. Although accuracy decreases slightly, the model becomes more effective for real-world churn prediction.

COMPARISON TABLE:

Metric	Logistic Regression	XGBoost (Tuned)
Accuracy	0.7420	0.7620
F1 Score	0.5000	0.5795
Recall	0.6339	0.8059
Precision	0.4128	0.4524
ROC-AUC	0.7791	0.8655

KEY INSIGHTS:

- XGBoost outperforms Logistic Regression across all metrics
- Significant improvement in Recall, which is critical for churn prediction
- Higher F1 Score indicates better balance between precision and recall
- ROC-AUC improvement confirms stronger predictive capability
- Slight drop in accuracy after tuning is acceptable due to improved churn detection

WHY RECALL IS IMPORTANT IN THIS PROBLEM:

In churn prediction, missing a churn customer (False Negative) can lead to revenue loss.

Therefore:

- High Recall ensures more churn customers are identified
- Precision is controlled to avoid excessive false alerts
- F1 Score is used as a balanced metric

FINAL MODEL SELECTION:

Based on overall performance and business relevance:

XGBoost with tuned threshold (0.42) is selected as the final model

CONCLUSION

The comparison clearly demonstrates that XGBoost, combined with proper imbalance handling and threshold tuning, provides a more reliable and effective solution for churn prediction compared to Logistic Regression. The model achieves a strong balance between recall and precision, making it suitable for real-world deployment.